



# The Economics of Bicycles for the Mind

Agrawal, Gans & Goldfarb (2025) — NBER Working Paper 34034

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Artificial Intelligence and Economic Modeling — Repository 2

<https://github.com/amchavezu/ai-02-agrawal>

# 1. A tool changes implementation, not the objective

## Question

How do cognitive tools change effort, productivity, and the value of human skills?

## Single mechanism

The tool directly improves implementation, lowering the marginal return to human implementation effort; judgment still selects payoffs and opportunities.

**Skills:** implementation  $s$ , payoff judgment  $\alpha$ , and opportunity judgment  $\{\gamma(t)\}$ .

Source: Agrawal, Gans & Goldfarb (2025), Sections 2–3.

## Conditional agent problem

$$\max_{e_t \geq 0} \underbrace{p(se_t; \theta)\alpha\Delta - c(e_t; \theta)}_{M(e_t; \theta)}.$$

The choice variable is effort  $e_t$ ; the constraint is  $e_t \geq 0$ .

At an interior optimum,

$$p'(se_t^*; \theta)s\alpha\Delta = c'(e_t^*; \theta).$$

Opportunity judgment changes how often the problem occurs, not this conditional FOC.

## 2. Proposition 1: more value with less effort

### All conditions used for strict conclusions

- ▶  $p$  increasing and weakly concave;  $c$  increasing and weakly convex.
- ▶  $s > 0$ ,  $\alpha\Delta > 0$ , and a finite positive opportunity multiplier  $\Gamma$ .
- ▶ Interior, unique optima; at least one relevant level improvement is strict.
- ▶ For every feasible  $e$ :  
 $p(se; 1) \geq p(se; 0)$ ,  $c(e; 1) \leq c(e; 0)$ ,  
$$\frac{p'(se; 1)}{c'(e; 1)} < \frac{p'(se; 0)}{c'(e; 0)} \quad (e > 0).$$

### Result

$$e_t^*(1) < e_t^*(0)$$

$$e_t^*(\theta) = e^*(\theta) \quad \forall t$$

$$V_0(1) > V_0(0)$$

The tool raises attainable net value but lowers the marginal payoff to direct implementation effort. Conditional choices are time-invariant because the one-opportunity objective is time-invariant.

At a corner or without strictness, the safe comparative statics are weak. Source: Proposition 1 and Appendix A.1.

### 3. Dependence, not marginals, governs the variance effect

$$V(\theta) = H + \theta Z, \quad H = \frac{\Delta^2}{4} \Gamma \alpha^2 s, \quad Z = \frac{\Gamma}{s},$$

$$\text{Var}(V) = \text{Var}(H) + 2\theta \text{Cov}(H, Z) + \theta^2 \text{Var}(Z),$$

$$\left. \frac{d \text{Var}(V)}{d\theta} \right|_0 < 0 \iff \mathbb{E}[\Gamma^2] < \mathbb{E}[\Gamma s] \mathbb{E}[\Gamma/s].$$

| Scenario                   | Condition/gap | Slope at 0 | $\theta^*$ or minimum |
|----------------------------|---------------|------------|-----------------------|
| Independent counterexample | (30) holds    | −0.1646    | 1.792 formal          |
| Correlated A               | −0.02762      | −0.3182    | 0.1403                |
| Correlated B               | +0.04616      | +0.5317    | boundary 0            |

Our student-chosen, standard-library extension checks 75 grid rows: direct variance equals the quadratic formula; reruns have identical hashes; and correlated A and B share both marginals.

## 4. I did not believe the endpoint claim

**Loose AI answer:** *Variance is U-shaped in AI quality.*

### Audit

$$\text{Var}'(V; \theta) = 2 \text{Cov}(H, Z) + 2\theta \text{Var}(Z),$$

$$\text{Var}'(V; 1) > 0 \iff \boxed{\theta^* < 1}.$$

Condition (30) proves  $\text{Var}'(V; 0) < 0$  and  $\theta^* > 0$ , but not the endpoint inequality.

**Counterexample:** (30) holds, but  $\theta^* = 1.792$  and the slope at one is  $-0.0727$ .

**Verdict:** the unconditional AI statement is incorrect; the paper's endpoint claim is under-conditioned. The object is  $\text{Var}_i(V_i(\theta))$ .

### My handwritten check



Pages 1–4: covariance, endpoint, counterexample, and corner.